Article

Environmental signal in the evolutionary diversification of bird skeletons

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Characterizing how variation in the tempo and mode of evolution has structured the phenotypic diversity of extant species is a central goal of macroevolution[1–3]. However, studies are typically limited to a handful of traits[4–6], providing incomplete information. We analyse morphological diversification in living birds, an ecologically diverse group[7], documenting structural scales from 'pan-skeletal' proportions down to the localized three-dimensional shape changes of individual bones. We find substantial variation in evolutionary modes among avian subgroups and among skeletal parts, indicating widespread mosaicism and possible differences in the structure of the macroevolutionary landscape across Earth's main environments. Water-linked groups, especially Aequorlitornithes (waterbirds), have repeatedly explored a large portion of their total morphospace, emphasizing variation in body proportions and in the shape of bones close to the body core, which are functionally related to the mechanics of locomotion[8]. By contrast, landbirds (Inopinaves) evolved distinct, group-specific body forms early in the aftermath of the K-Pg and subsequently emphasized local shape variation, especially in the head and distal limb bones, which interact more directly with the environment. Passerines, which comprise more than half of all bird species, show a conservative evolutionary dynamic that resulted in low disparity across all skeletal parts. Evidence for early establishment of the morphospace of living birds is clear for some skeletal parts, including beaks and the combined skeletal morphology. However, we find little evidence for early partitioning of that morphospace, contrary to more specific predictions of 'niche-filling' models[1,9]. Nevertheless, early divergence among broad environmental types may have caused an early divergence of evolutionary modes, suggesting an important role for environmental divergence in structuring the radiation of crown-group birds.

Organisms show a spectacular phenotypic disparity, and this disparity is distributed unevenly among lineages and environments. Macroevolutionary research has long sought to understand how this evolved, and the processes that cause some groups to attain far greater disparity than others[1,2]. Individuals, comprising many anatomical parts, are the subject of natural selection. However, even the most comprehensive macroevolutionary studies have used data from only limited parts of the organism, especially the head (for example, ref. [4–6]), leaving important questions unanswered. For example, do different structures show distinct evolutionary trajectories (that is, mosaic evolution)? To what extent do evolving lineages express differing amounts and kinds of phenotypic variation (that is, evolutionary versatility)? And to what extent are patterns of morphological evolution shaped by interactions among species, versus environmental factors? Environmental and among-species drivers are conceptually unified under the widespread 'niche-filling' narrative, in which environment controls the availability of ecological niches and interactions among species drive partitioning of those niches, with direct effects on morphological evolution[1,9–11]. This hypothesis predicts early establishment of a broad phenotypic space during evolutionary radiations into new adaptive zones (for example, ref. [12]), as well as early partitioning of that space into subgroup-specific subspaces (for example, ref. [13]).

Birds are ecologically diverse and speciose, with prominent ecological roles in practically all Earth's surface environments[7], providing an informative study system for these questions. As vertebrates, they are structurally complex and major hypotheses propose a highly modular pattern of bird evolution[14–16]. Molecular phylogenetics, including genome-scale analyses, provide an increasingly clear understanding of avian relationships and divergence times[17,18]. The latest common ancestor of all extant bird species was probably a ground-dwelling species[19,20] and lived in the Late Cretaceous (83–66 million years ago (Ma)). Extant bird diversity is suggested, at least in part, to have resulted from an ecological radiation following the Cretaceous-Paleogene (K-Pg) mass extinction event 66 Ma (for example, refs. [21,22]). This radiation was characterized by early partitioning of lineages among environments, giving rise to speciose water- and land-based groups, as also seen during earlier divergences (for example, arboreal enantiornithines versus water-linked ornithuromorphs in the Mesozoic[23]). This divergence

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along environmental lines is embodied most clearly by the ecologically disparate sister clades Aequorlithornithes (waterbirds) and Inopinaves (landbirds), the latter of which includes the hyper-diverse Passeriformes (passerines)[17]. Some of these groups have conserved their climatic niches since the early Cenozoic suggesting early partitioning of environmental space was central to the crown avian diversification[24].

In addition to early partitioning of environmental space, many hypotheses of bird diversification also assume a pattern of early partitioning of morphological diversity after the K-Pg (for example, refs. [16,25–27]), perhaps similar to the post-K-Pg radiation of crown placental mammals[1,28,29]. However, strong evidence for these patterns remains elusive and mostly comes from characterizations of variation for single anatomical structures, with especially intensive study of head[5,30–32] and beak[4,33–35] morphology, yielding mixed results[4–6,30].

Here, we take an anatomically integrated approach to avian macroevolution, examining different aspects of morphological variation across the skeleton of birds, asking how they have varied through time and among groups with different ecologies. The vertebrate skeleton is a structural scaffold for most biological systems, with demonstrated links to ecology and well-understood functional importance[8]. We quantified skeletal variation in the three-dimensional (3D) shape and size of 13 skeletal elements in 228 species spanning the entire phylogeny of extant birds. This dataset provides information spanning from the Late Cretaceous origin of the bird crown group up to the present, including the deep divergences between main avian subgroups.

We analyse several hierarchical levels of skeletal structure: (1) the relative sizes of elements drawn from across the skeleton, combined to reflect body proportions; (2) the unified relative sizes and shapes (='form') of skeletal elements, combined to reflect overall skeletal form; (3) the shapes of individual skeletal elements (size-normalized) and (4) local shape variation of elements, both individually and when combined together into regions or the whole skeleton (proportions-normalized, Methods). This allows us to analyse the relative contributions of variation at different structural scales, offering insights beyond the scope of previous studies. For example, local shape variation can be swamped by proportional shape variation in conventional analyses. Nevertheless, it has potentially high functional importance, encompassing changes in joint surface morphologies and processes for muscle attachment, which we analytically separate from proportional variation such as elongation, widening or deepening (Methods).

We assess variation in morphological diversity (disparity) among structural scales, anatomical regions and groups, evaluating hypotheses about the evolutionary mode by which that disparity was acquired through time. Our analyses differentiate between two separate predictions of the niche-filling model of morphological diversification, using comparisons to uniform, multivariate Brownian motion models to establish statistical significance: (1) the hypothesis of 'early establishment' of a large portion of the morphospace of extant lineages; and (2) the hypothesis of 'early partitioning' of that morphospace, in which subgroups divide morphospace into group-specific subspaces that restrict subsequent evolution (Methods, Extended Data Figs. 1 and 2, Methods summary and also ref. [11]). Throughout our analyses, we find great variation in the mode of morphological evolution among elements and regions of the skeleton (Figs. 1–4). This confirms the widespread occurrence of mosaic evolution in birds, and we suggest that it is most probably ubiquitous among anatomically complex organisms, necessitating a more detailed dissection of the patterns shown by different anatomical regions.

We find that birds established a wide morphospace of overall skeletal form during their early evolution (Fig. 1), driven primarily by divergence of the head and forelimb morphology (Supplementary Fig. 15). This pattern is also evident for some individual elements, including the skull and mandible (Supplementary Fig. 16), as well as the local shape variation of the sternum and tarsometatarsus, and overall shape (but not local shape) of the humerus, radius and ulna (Supplementary Fig. 16). Other elements, especially of the hindlimb and pectoral girdle, show no evidence of this (Supplementary Fig. 15). Our analysis of early establishment of bird beak morphospace directly replicates most of the pattern from Cooney et al.[4], when we exactly replicate their approach, even though they used a much larger dataset (2,000 species), when we exactly replicate their approach (Fig. 1b). This shows that our dataset, in spite of its smaller size, has sufficient power to test hypotheses of early morphological evolution among birds, which is the focus of our analysis and interpretations. Our finding of early establishment of the morphospace of overall skeletal form, the head, forelimb and various individual elements of birds is consistent with fossil evidence that some groups established their distinctive morphologies by the early Cenozoic (for example, penguins, rollers, mousebirds, frigatebirds[36]). It also complements palaeontological evidence that many other animal groups also established a large portion of their total disparity during their early history[12,37].

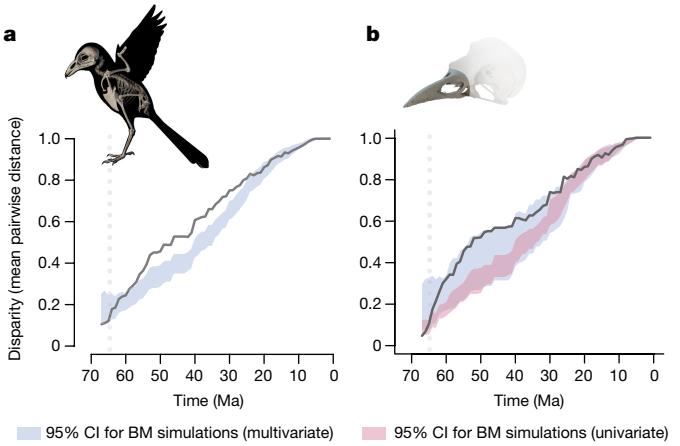


Fig. 1 | Early establishment of the morphospace for skeletal form and beak shape during the diversification of Neornithes. a,**b**, Skeletal form (**a**) and beak shape (**b**). Empirical disparity time series based on ancestral character estimation for Neornithes (grey, continuous line) is compared to expectation under a set of 100 simulations under diffusive, multivariate Brownian motion (BM) model (blue shaded area indicating 95% confidence intervals (CI); red shaded area for univariate simulations for beak shape as in ref. [4]). Timing of the K-Pg mass extinction event is indicated by the dotted vertical line. Analyses of other elements and anatomical regions across all our data types are shown in Supplementary Figs. 15 and 16.

In spite of the early establishment of many aspects of bird morphospace, we find little evidence for strong early partitioning of that morphospace (Figs. 2 and 3 and Extended Data Figs. 3–10). We quantify this using a version of the morphological diversity index (MDI) modified to accommodate high dimensional data (MDI[11], Extended Data Figs. 1 and 2). Negative MDI values indicate a pattern of early partitioning, whereby subclades explore smaller morphospaces than expected given the disparity of the full clade. This would indicate that whole-clade morphospace was divided into group-specific subspaces early on (Extended Data Fig. 2 ref. [11]), as predicted by the niche-filling hypothesis[1,9]. However, we instead find mostly positive MDI values, indicating that subclade disparities are generally higher than expected given the disparity of the full clade, indicating that constraints on morphological evolution at the level of all birds are similar to those of its subgroups (Extended Data Fig. 2). This result indicates only weak partitioning of avian skeletal space, allowing at least some groups to engage in repeated re-exploration of morphospace throughout much of avian evolution (Figs. 2 and 3; Neornithes), especially among waterbirds (below). This pattern is evident both for the combined form of skeletal regions and for the individual shapes of many bones, both across birds and in most subgroups (Fig. 3 and Extended Data Fig. 3).

Various bird groups partitioned surface environments following the K-Pg mass extinction. The most speciose of these are the sister clades

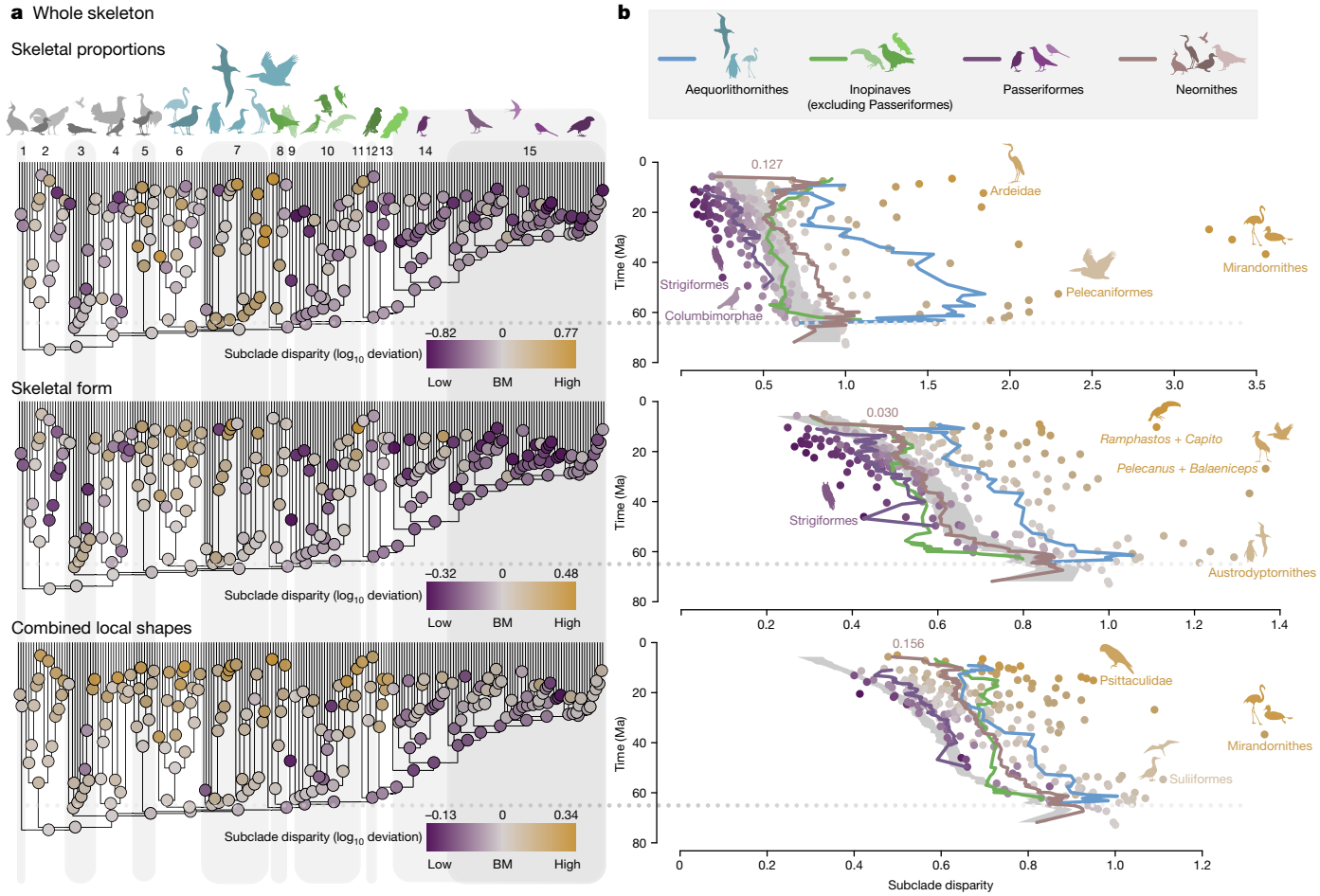


Fig. 2 | Whole skeletal subclade disparity–age plotted across avian phylogeny and through time. a, Whole skeleton subclade disparity values colour-coded on the basis of difference from expectations under a constant diffusive model of evolution (Brownian motion, BM), plotted at the basal node of each clade in the phylogeny for three kinds of morphological data: skeletal proportions (top), combined overall form for the whole skeleton (middle, combined element shapes after correcting for body mass, plus relative sizes among elements) and local shapes from all skeletal elements combined together (proportions free combined shapes; bottom, combined shapes of elements corrected for body mass and main axes proportional changes in shape, Methods). Subclade disparity values are $\log_{10}$-transformed deviations of empirical values from means of 200 simulations of morphological evolution under BM. Purples represent lower disparities than expected under BM for that clade, ochres represent higher disparities than expected under BM. **b**, Patterns of mean binned subclade disparity against subclade age for all birds (Neornithes) and three target lineages. Only the empirical curved of Neornithes (brown) relates to the null expectations for BM, the three subclade curves (Aequorlithornithes, Inopinaves and Passeriformes) are only plotted for reference (for similar curves for other subclades, see Extended Data Figs. 5–10). Grey envelope represents the maximum and minimum values for each bin of the 200 simulations of morphological evolution under BM. Individual colour-coded clade values are plotted and some extreme values are labelled. Time scale in million years apply for both parts of the figure and the dashed horizontal line at 67 Ma indicates the timing of K-Pg extinction. Clades in **a** indicated with the following numbers: 1 Palaeognathae, 2 Galloanserae, 3 Strisores, 4 Columbaves, 5 Gruiformes, 6 Mirandornithes + Charadriiformes, 7 Ardeae, 8 Accipitrimorphae, 9 Strigiformes, 10 Eucavitaves, 11 Cariamiformes, 12 Falconiformes, 13 Psittaciformes, 14 Suboscines (Passeriformes) and 15 Oscines (Passeriformes).

landbirds (Inopinaves/Telluraves + Opisthocomiformes) largely radiating into terrestrial/arboreal niches, and waterbirds (Aequorlitornithes/Ardeae + Charadriiformes + Mirandornithes) largely diverging in marine/coastal/estuarine environments (with rare reversions; Supplementary Fig. 35). We have shown that this environmental divergence did not produce the expected early partitioning of morphospace. Nevertheless, we do find strong evidence that this split gave rise to stark differences in overall disparity (Figs. 2 and 4), and in the evolutionary mode by which this disparity was obtained (Fig. 3 and Extended Data Figs. 3–10). Landbirds and their subgroups, especially passerines (Passeriformes), show low overall disparity, with few exceptions (for example, high disparity of skeletal form in toucans + barbets; Figs. 2 and 4). Unlike other groups, landbirds also show early partitioning of overall skeletal form (significantly negative MDI), although this is not evident for body proportions or for individual bone morphologies (except the mandible; Fig. 2 and Extended Data Figs. 3, 6 and 9). By contrast, waterbirds show overall high disparity, particularly for skeletal proportions and for overall skeletal form (Figs. 2–4; also found when ardeans and charadriiforms are analysed separately, Supplementary Fig. 34) and show strong evidence for a lack of partitioning of morphospace in the evolution of skeletal form and proportions (significantly positive MDI; Fig. 3).

Similar patterns of relative disparity are obtained for other water-linked groups when compared to closely related terrestrial groups. This includes comparisons of gruiforms (rails, finfoots, flufftails, limpkins, trumpeters and cranes) to non-landbird terrestrial neoavians (a diverse assemblage including nightjars, swifts, hummingbirds, pigeons, bustards, cuckoos among others), and of anseriforms (ducks and geese) to galliforms (landfowl) (Fig. 2, Extended Data Fig. 4 and Supplementary Figs. 35 and 36). Although sample sizes are too small to rigorously investigate MDI, these water-linked groups show

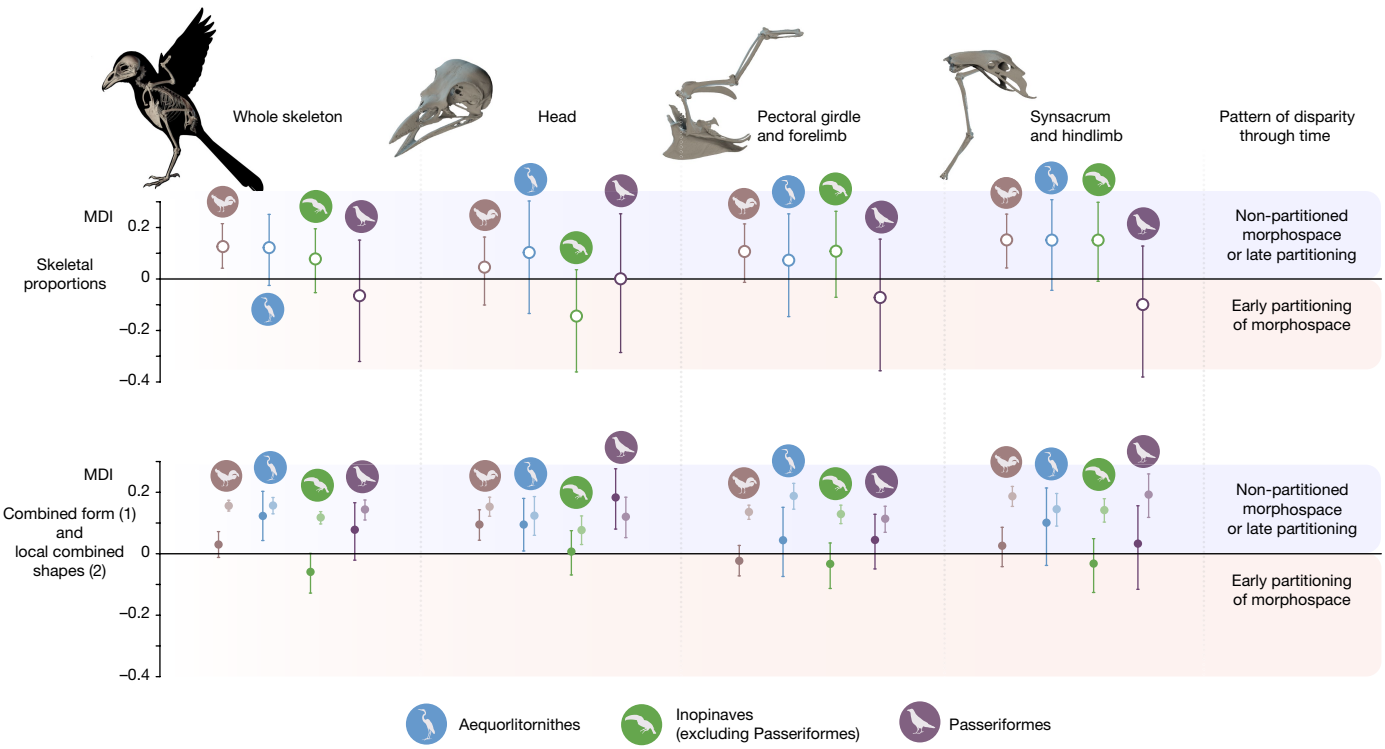


Fig. 3 | Tests of early partitioning of avian skeletal space using the MDI. MDI means and confidence intervals are derived from comparison of subclade disparities to simulations under Brownian motion (Methods), shown here for all aspects of morphological variation of the whole skeleton and the three main anatomical regions in all Neornithes, and in Aequorlithornites, Inopinaves and Passeriformes (see Extended Data Fig. 2 for the patterns of individual skeletal elements). Mean MDI values for each partition, lineage and/or data type are represented by a dot and associated 95% confidence intervals are represented as whiskers. See Extended Data Fig. 2 for a detailed account of the idealized patterns associated with each mode of evolution of disparity through time.

significantly greater disparity in skeletal proportions than their terrestrial relatives, consistent with an association of broad environmental class with skeletal disparity in birds (Extended Data Fig. 4).

We find that non-passerine landbirds are not simply less disparate or more constrained than waterbirds across all structural scales, contrary to their low disparity of skeletal proportions and most element shapes. Local shape variation was especially important to landbird diversification: differences between the disparities of waterbirds and non-passerine landbirds are substantially reduced when analysing local shape variation (Figs. 2 and 4). Comparison of within-group disparity to expectations under a uniform model of morphological diversification for all birds (Methods and Extended Data Fig. 1) indicates that landbirds collectively have emphasized variation in the skeletal elements that interact most directly with the environment (skull/mandible, carpometacarpus, tarsometatarsus) (Fig. 4). This is more prominent in some subgroups; for instance, piciiforms (toucans, woodpeckers and kin) show large variation in skull and mandible shapes whereas psittaciforms (parrots and cockatoos) demonstrate large variation in the local aspects of the morphology of the mandible and the tarsometatarsus, including articular facets of trochleae, with implications for foot function (Supplementary Figs. 1 and 2). Conversely, waterbirds show higher shape disparity in elements closer to the body core (sternum, coracoid, femur, tibiotarsus) (Fig. 4), which are functionally related with the lever mechanics of the wingstroke and stride[8,38]. Exceptions to these generalizations do exist. For example, important divergences of other elements did occur within waterbird lineages (for example, extreme skull and mandible shape divergence between grebes and flamingos, Supplementary Figs. 21 and 22), and strisoreans (nightjars, frogmouths, swifts, hummingbirds) show a hyper-variable forelimb morphology contrasting with their extremely conservative hindlimb variation (Extended Data Fig. 4 and Supplementary Figs. 19 and 20). These observations illustrate a dazzling complexity of skeletal evolvability across the living radiation of birds (see Supplementary Figs. 1–34 for more detailed patterns of morphological evolution in other lineages).

Passerines, a landbird subclade that includes more than half of modern bird species, show noticeably low disparity in all studied aspects of morphology, including local shape variation, and even when accounting for their relatively recent origins using a uniform model of morpho-logical diversification (Figs. 2 and 4). This indicates that the low disparity of passerine morphology most probably results from a conservative evolutionary dynamic compared to other birds, and not from a lack of time to diversify. Therefore, morphological divergence was less important during the passerine super-radiation than in other birds. Instead, passerine diversification may be more related to other factors such as geographic differentiation (for example, ref. [39]), behavioural divergence, subtle ecomorphological differences related to finer-scale habitat partitioning (for example, ref. [40]) or variation in display signals such as plumage patterns or song (for example, refs. [41,42]).

Our results indicate remarkable variation in the dynamics of phenotypic evolution among main avian subgroups, suggesting fundamental differences in the structure of the macroevolutionary landscape experienced by birds evolving in water-related ecosystems compared to those in arboreal/terrestrial environments.

In summary, we found that landbirds established group-specific body forms related to locomotor and foraging strategies during their early history, documenting early partitioning of avian morphospace by a major terrestrial subgroup. However, this pattern is unique to the combined skeletal form of landbirds, and not typical of patterns across

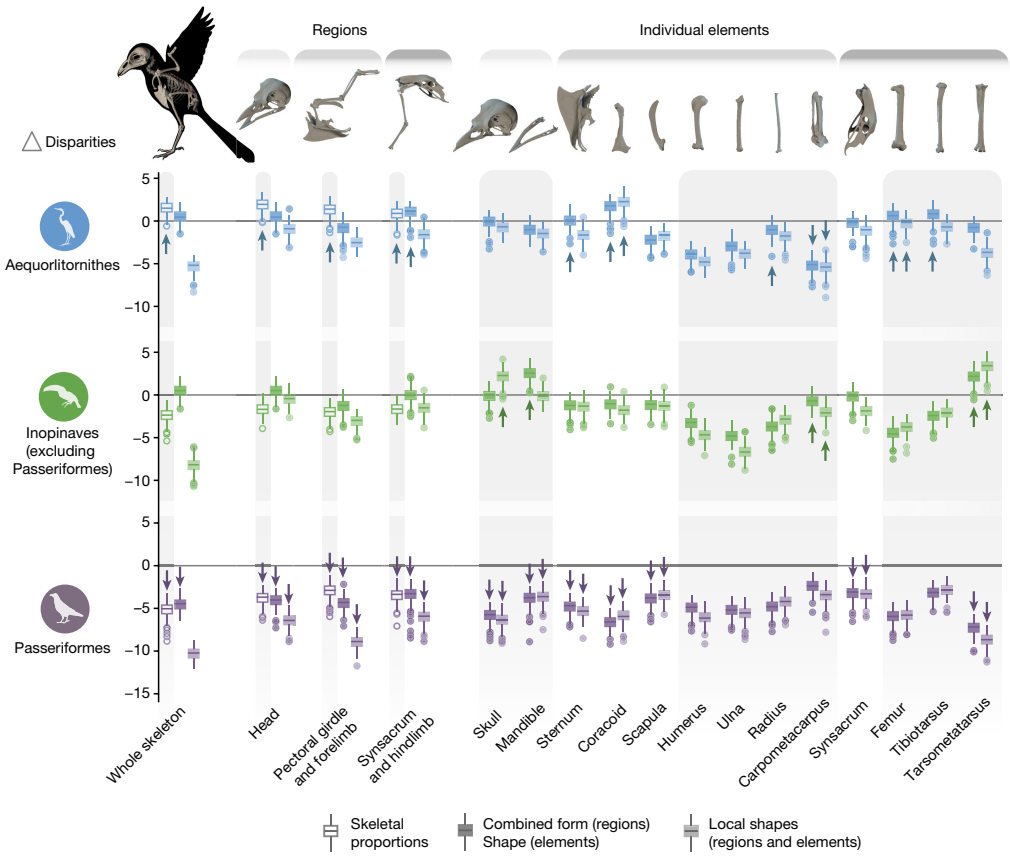


Fig. 4 | Deviations of group-specific disparities from the disparity expected if all birds were evolving under a uniform Brownian motion model of evolution. Boxplots summarize the distribution of values, calculated as empirical disparities minus disparities for the 200 simulated values for Aequorlithornithes, Inopinaves and Passeriformes. High values indicate higher than expected disparity and low values indicate lower than expected disparity. All partitions and data types are shown, namely, the whole skeleton, three main skeletal regions and individual bones and the four different aspects of morphological variation. Arrows highlight the lineage, partition and/or data type that shows particularly high (upwards pointing arrows) or low (downwards pointing arrows) values. Values are normalized by interquartile range.

birds. Landbirds subsequently emphasized lower-scale changes in the skull, mandible, carpometacarpus and tarsometatarsus, terminal elements that show some of the strongest correlations with ecology across the bird skeleton[32]. For instance, variation in tarsometatarsus structure encodes important variation in the foot-use traits that vary markedly among landbirds, including digit orientations, arboreality and raptoriality[43,44], and variation in skull and mandible shapes encodes dietary variation, although these relationships are complex[32,35,45]. One possible explanation for these patterns is that land-based environments, especially arboreal settings, are topologically complex and offer substantial opportunities for fine-grain niche partitioning among similar ecologies.

By contrast, waterbirds continually re-explored a larger space of body proportions and body core elements, and high disparity of body proportions is also seen in other water-linked groups (anseriforms, gruiforms). This suggests much greater evolutionary versatility of coarse-grained locomotory adaptations and greater discontinuities between water-related adaptive zones. For example, the divergence between penguins (wing-propelled divers) and tubenoses (mostly dynamic soarers) indicates a complete re-organization of the muscle function in the trunk, fore- and hindlimbs[38], which in fact occurred a second time during the origins of diving petrels (also wing-propelled divers) from within tubenoses, and in parallel within shorebirds, with the origin of wing-propelled diving in auks[7].

Our findings resonate with a hypothesis of 'interzonal' diversification in waterbirds and early landbirds (that is, diversification between widely distinct adaptive zones) versus 'subzonal' diversification in the subsequent evolution of landbirds (that is, bounded diversification within adaptive zones). These modes of diversification were proposed qualitatively, and in the absence of a detailed knowledge of avian phylogeny more than 65 years ago[1]. 'Subzonal' diversification, possibly facilitated by behavioural differentiation, may have been even more important for the diversification of passerines, providing one potential explanation for the apparent paradox of high species diversity coupled with low morphological disparity. Differences in developmental constraints may also explain these patterns, and are not mutually exclusive of environmental explanations, but remain poorly understood.

Morphological macroevolution seeks to understand the processes that gave rise to the phenomenal disparity of extant species, encompassing their full phenotypic diversity. However, many previous studies have focussed on just specific aspects of phenotype, usually those with assumed links to particular ecological traits (for example, refs. [4,6,41] although see ref. [46]). We instead quantified phenotypic variation across much of the skeleton, permitting an unprecedented characterization of macroevolutionary patterns at several structural scales. Our results indicate that, even when early establishment of a wide morphospace is evident, associated with early partitioning of environmental space (as with deep divergences between water- and land/arboreal-related bird groups) this did not give rise to early partitioning of morphological traits.

Our findings therefore indicate a decoupling of morphospace partitioning from the partitioning of environmental space in crown-group birds, for which hypotheses have yet to be proposed. This potentially could reflect the importance of finer-scale ecological partitioning as a driver of morphological innovation during later avian evolution. Alternatively, patterns of morphological partitioning may not faithfully

reflect underlying patterns of ecological partitioning given that morphology and ecology are only weakly correlated at large phylogenetic scales in birds[32,35,45]. Furthermore, although many studies have assumed that ecological partitioning occurs mainly during the early evolution of groups (for example, this is implicit in the 'early burst' model of morphological evolution[47]), ecological shifts may continue to occur during later evolution, given that turnover events are common throughout the histories of most main groups[47]. Directly evaluating patterns of ecological evolution is therefore important but may be challenging because of difficulties fully quantifying the niches of organisms. Nevertheless, our observations question the widespread assumption of 'niche-filling' models, that saturation of ecological space occurs during the course of an evolutionary radiation and causes early partitioning of morphological space[9,10,48].

We suggest instead that evolutionary shifts between main ecological types have structured the phenotypic disparity of organisms by altering the dynamics of morphological evolution: as evidenced by the very different patterns seen in waterbirds and landbirds and replicated to a certain extent between other water-linked and terrestrial sister lineages in our sample (Extended Data Fig. 4 and Supplementary Figs. 17–36). Our findings emphasize the potential for environmental factors to influence phenotypic macroevolution in a complex manner, beyond simply defining the niches available to evolving lineages, and resonate with similar contrasting patterns found between marine and fresh-water teleostean fishes[49]. We propose a potential role for the physical or ecological structure of environments in determining evolutionary variation among units of the avian skeleton and at different structural scales. It is also possible that differences in spatial and climatic heterogeneity between water- and land- or arboreal-based environments, either through space or in time, may explain emergence of interzonal versus subzonal patterns in different groups of ancestrally terrestrial organisms. Incorporating these insights into macroevolutionary models and theory will be necessary to achieve a deeper understanding of evolutionary processes that control the distribution of disparity in space and time, in particular, to deepen in our understanding of the relationship among ecological opportunity and the diversification of species and their phenotypes.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-022-05372-y.

1. Simpson, G. G. *The Major Features of Evolution* (Columbia Univ. Press, 1953).
2. Gould, S. J. *Ontogeny and Phylogeny* (Harvard Univ. Press, 1985).
3. Jablonski, D. Approaches to macroevolution: 2. Sorting of variation, some overarching issues, and general conclusions. *Evol. Biol.* **44**, 451–475 (2017).
4. Cooney, C. R. et al. Mega-evolutionary dynamics of the adaptive radiation of birds. *Nature* **542**, 344–347 (2017).
5. Felice, R. N. & Goswami, A. Developmental origins of mosaic evolution in the avian cranium. *Proc. Natl Acad. Sci. USA* **115**, 201716437 (2018).
6. Navalón, G., Marugán-Lobón, J., Bright, J. A., Cooney, C. R. & Rayfield, E. J. The consequences of craniofacial integration for the adaptive radiations of Darwin's finches and Hawaiian honeycreepers. *Nat. Ecol. Evol.* **4**, 270–278 (2020).
7. Gill, F. B. *Ornithology* (Macmillan, 1995).
8. Kardong, K. V. *Vertebrates: Comparative Anatomy, Function, Evolution* (Heinle and Heinle Publishers, 1997).
9. Schluter, D. *The Ecology of Adaptive Radiation* (Oxford Univ. Press, 2000).
10. Losos, J. B. Adaptive radiation, ecological opportunity, and evolutionary determinism: American Society of Naturalists EO Wilson Award address. *The American Naturalist* **175**, 623–639 (2010).
11. Harmon, L. J., Schulte, J. A., Larson, A. & Losos, J. B. Tempo and mode of evolutionary radiation in iguanian lizards. *Science* **301**, 961–964 (2003).
12. Hughes, M., Gerber, S. & Wills, M. A. Clades reach highest morphological disparity early in their evolution. *Proc. Natl Acad. Sci. USA* **110**, 13875–13879 (2013).
13. Ackerly, D., Schwilk, D. & Webb, C. Niche evolution and adaptive radiation: testing the order of trait divergence. *Ecology* **87**, S50–S61 (2006).
14. Gatesy, S. M. & Dial, K. P. Locomotor modules and the evolution of avian flight. *Evolution* **50**, 331–340 (1996).
15. Dececchi, T. A. & Larsson, H. C. Body and limb size dissociation at the origin of birds: uncoupling allometric constraints across a macroevolutionary transition. *Evolution* **67**, 2741–2752 (2013).
16. Clarke, J. A. & Middleton, K. M. Mosaicism, modules, and the evolution of birds: results from a Bayesian approach to the study of morphological evolution using discrete character data. *Syst. Biol.* **57**, 185–201 (2008).
17. Prum, R. O. et al. A comprehensive phylogeny of birds (Aves) using targeted next-generation DNA sequencing. *Nature* **526**, 569–573 (2015).
18. Oliveros, C. H. et al. Earth history and the passerine superradiation. *Proc. Natl Acad. Sci. USA* **116**, 7916–7925 (2019).
19. Field, D. J. et al. Early evolution of modern birds structured by global forest collapse at the end-Cretaceous mass extinction. *Curr. Biol.* **28**, 1825–1831. e1822 (2018).
20. Benito, J. et al. 40 new specimens of Ichthyornis provide unprecedented insight into the postcranial morphology of crownward stem group birds. *PeerJ* **10**, e13919 (2022).
21. Field, D. J. et al. Timing the extant avian radiation: the rise of modern birds, and the importance of modeling molecular rate variation. *Bull. Am. Mus. Nat. Hist.* **440**, 159–181 (2020).
22. Feduccia, A. 'Big bang' for tertiary birds? *Trends Ecol. Evol.* **18**, 172–176 (2003).
23. Chiappe, L. M. & Qingjin, M. *Birds of Stone: Chinese Avian Fossils from the Age of Dinosaurs* (JHU Press, 2016).
24. Saupe, E. E. et al. Climatic shifts drove major contractions in avian latitudinal distributions throughout the Cenozoic. *Proc. Natl Acad. Sci. USA* **116**, 12895–12900 (2019).
25. Nudds, R., Dyke, G. & Rayner, J. Forelimb proportions and the evolutionary radiation of Neornithes. *Proc. R. Soc. London Ser. B Biol. Sci.* **271**, S324–S327 (2004).
26. Wang, X. & Clarke, J. A. Phylogeny and forelimb disparity in waterbirds. *Evolution* **68**, 2847–2860 (2014).
27. Crouch, N. M. & Ricklefs, R. E. Speciation rate is independent of the rate of evolution of morphological size, shape, and absolute morphological specialization in a large clade of birds. *Am. Naturalist* **193**, E78–E91 (2019).
28. Brocklehurst, N., Panciroli, E., Benevento, G. L. & Benson, R. B. Mammaliaform extinctions as a driver of the morphological radiation of Cenozoic mammals. *Curr. Biol.* **31**, 2955–2963.e4 (2021).
29. Osborn, H. F. The law of adaptive radiation. *Am. Naturalist* **36**, 353–363 (1902).
30. Felice, R. N. et al. Decelerated dinosaur skull evolution with the origin of birds. *PLoS Biol.* **18**, e3000801 (2020).
31. Felice, R. N., Tobias, J. A., Pigot, A. L. & Goswami, A. Dietary niche and the evolution of cranial morphology in birds. *Proc. R. Soc. B* **286**, 20182677 (2019).
32. Orkney, A., Bjarnason, A., Tronrud, B. C. & Benson, R. B. Patterns of skeletal integration in birds reveal that adaptation of element shapes enables coordinated evolution between anatomical modules. *N. Ecol. Evol.* **5**, 1250–1258 (2021).
33. Chira, A. M. & Thomas, G. H. The impact of rate heterogeneity on inference of phylogenetic models of trait evolution. *J. Evol. Biol.* **29**, 2502–2518 (2016).
34. Rombaut, L. M. et al. Allometric conservatism in the evolution of bird beaks. *Evolution Lett.* **6**, 83–91 (2022).
35. Navalón, G., Bright, J. A., Marugán-Lobón, J. & Rayfield, E. J. The evolutionary relationship among beak shape, mechanical advantage, and feeding ecology in modern birds. *Evolution* **73**, 422–435 (2019).
36. Mayr, G. *Avian Evolution: The Fossil Record of Birds and its Paleobiological Significance* (John Wiley & Sons, 2016).
37. Foote, M. The evolution of morphological diversity. *Ann. Rev. Ecol. Systematics* **28**, 129–152 (1997).
38. Bribiesca-Contreras, F., Parslew, B. & Sellers, W. I. Functional morphology of the forelimb musculature reflects flight and foraging styles in aquatic birds. *J. Ornithol.* **162**, 779–793 (2021).
39. Phillimore, A. B. et al. Sympatric speciation in birds is rare: insights from range data and simulations. *Am. Naturalist* **171**, 646–657 (2008).
40. MacArthur, R. H. Population ecology of some warblers of northeastern coniferous forests. *Ecology* **39**, 599–619 (1958).
41. Cooney, C. R. et al. Sexual selection predicts the rate and direction of colour divergence in a large avian radiation. *Nat. Commun.* **10**, 1773 (2019).
42. Tobias, J. A. et al. Species coexistence and the dynamics of phenotypic evolution in adaptive radiation. *Nature* **506**, 359 (2014).
43. Del Hoyo, J. et al. *Handbook of the Birds of the World Alive* (Lynx Editions, 2017).
44. Botelho, J. F., Smith-Paredes, D. & Vargas, A. O. Altriciality and the evolution of toe orientation in birds. *Evolution. Biol.* **42**, 502–510 (2015).
45. Natale, R. & Slater, G. J. The effects of foraging ecology and allometry on avian skull shape vary across levels of phylogeny. https://doi.org/10.1086/720745 (2022).
46. Pigot, A. L. et al. Macroevolutionary convergence connects morphological form to ecological function in birds. *Nat. Ecol. Evol.* **4**, 230–239 (2020).
47. Slater, G. J. Iterative adaptive radiations of fossil canids show no evidence for diversity-dependent trait evolution. *Proc. Natl Acad. Sci. USA* **112**, 4897–4902 (2015).
48. Rabosky, D. L. & Lovette, I. J. Density-dependent diversification in North American wood warblers. *Proc. R. Soc. B. Biol. Sci.* **275**, 2363–2371 (2008).
49. Friedman, S., Collyer, M., Price, S. & Wainwright, P. Divergent processes drive parallel evolution in marine and freshwater fishes. *Syst. Biol.* **4**, syab080 (2021).

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Methods

Dataset, phylogenetic hypothesis and morphological variables

Our dataset comprises 3D morphometric data for 13 skeletal units in 228 species, providing a phylogenetically and ecologically broad sample of extant avian diversity (see Supplementary Table 1 for a full list of species and specimens). It is substantially expanded from the recent dataset of Orkney et al.[32], by adding 79 species, improving the representation of many groups, especially passerines. To illustrate that our sample of extant bird species (n = 228) provides an effective sample of ancient bird divergences, we compared the count of phylogenetic lineages through time for a supertree of 9,993 extant bird species sourced from ref. [4] (complete supertree from ref. [50], using the backbone from ref. [17]) to the count of lineages in a pruned tree that includes only those species that we sampled. Results show that our species sample achieves a near-complete coverage of deep bird divergences up to around 30 million years ago (Supplementary Fig. 37) and only missing out a significant portion of bird diversity (half the lineages) at about 20 million years ago onwards.

Digital skeletal models were segmented from high-resolution micro-CT (computed-tomography) scans of complete skeletons in Avizo 9.3 (Thermo Fisher Scientific, Zuse Institute). We placed landmarks and curve semilandmarks on these models, yielding a dense, quantitative representation of each element's geometry (Supplementary File 1: full landmark, for details on landmarking scheme, including consideration of homology, see ref. [51]), and thereby providing a broad representation skeletal form (shape and size) comparable to previous studies (for example, ref. [5]). Some elements of the skeleton, such as the pedal phalanges, were not landmarked for practical reasons (for example, availability in skeletal museum collections) despite their ecological relevance[52]. Our analyses were conducted in R v.4.0.2 (ref. [53]) using functions primarily from the packages ape v.5.5, geomorph v.4.0.1 (ref. [54]), RRPP v.0.6.2 (ref. [55]) and geiger v.2 (ref. [56]), as well as custom code (Supplementary Files 3 and 4). We use a composite phylogenetic framework combining recent phylogenetic hypotheses for non-passerines[17] and passerines[18] (see Supplementary File 2 for tree files and Supplementary Table 1 for a taxonomic match between source phylogenies and our morphological data).

Shape data (Procrustes residuals) were isolated from size, position and rotational information by applying generalized Procrustes analysis[57] to the 3D landmark coordinates of each element using the gpagen function from geomorph, sliding semilandmarks to minimize bending energy differences from the mean shape[58]. Common allometric information was factored out from element shapes by taking residuals from a phylogenetic generalized least squares (pGLS) regression of Procrustes residuals on $\log_{10}$-transformed body mass, using the function procD.pgls of geomorph. The 'allometry-free' residual shape matrix was added to the mean shape for use as 'shape' in subsequent analyses. Element size was indexed as the centroid size from our landmark/semilandmark constellations and relative element sizes were taken as the residuals from a pGLS regression of $\log_{10}$-element size on $\log_{10}$-transformed body mass. The 'form' of individual elements is represented by shape multiplied by relative size.

We analysed data representing structure at four hierarchical levels of organization: (1) overall skeletal form, as the combined relative sizes and allometry-corrected shapes (='forms') of all skeletal elements, or subsets representing the head (skull, mandible), pectoral limb (scapula, coracoid, sternum, humerus, ulna, radius, carpometacarpus) and pelvic limb (synsacrum, femur, tibia, tarsometatarsus); (2) overall skeletal proportions represented by a matrix containing the relative sizes of individual elements across the skeleton and for subsets (head, pectoral limb, pelvic limb); (3) the allometry-corrected shapes of individual skeletal elements and (4) local shape variation of elements, both individually and combined together per region, calculated as explained below. Combined configurations were generated by overlying landmark configurations for individual elements in the same space, centring them on the origin then rescaling the combined configuration to unit centroid size following recent recommendations for best practices[59], using custom code and the function gpagen.

Therefore, for individual elements we studied (1) allometry-free shape and (2) proportions-normalized shape, whereas for combined elements (whole skeleton, head, pectoral limb and pelvic limb) we studied (1) skeletal proportions, (2) allometry-free form (shape plus size) and (3) proportions-normalized combined shapes.

Local shape variation (proportion-normalized shape)

Local shape variation for each element was calculated by normalizing element shapes according to element proportions. This was done by taking the residuals from a pGLS regression of Procrustes residuals (shape) against $\log_{10}$ body mass and the maximum length, breadth and depth of each element, measured along standard anatomical axes, and subsequently called '*xyz* distances'. *xyz* distances were calculated as the distance between selected pairs of landmarks in the Procrustes residuals (Supplementary Table 1 for all distances and Supplementary Table 3 for all pairs of landmarks per axis and element). This new approach is intended to remove aspects of shape variation that are correlated only to overall stretching or squashing of the aspect ratios or proportional lengths of skeletal units (that is, elongation, widening or deepening of an anatomical structure), which may primarily be related to changes in overall skeletal proportions. The residual variation reflects local-scale shape variation. For example, variation in articular facets or processes for attachment of muscles or tendons that may hold important functional information.

For each skeletal element, we built additive pGLS models for all the combinations of *xyz* distances, analysed as covariates (for example, 'shape ~ $\log_{10}$ body mass + x + y'; shape roughly $\log_{10}$ body mass + x + y + z). We then retained as proportion-normalized shapes the residuals from the pGLS model that satisfies two conditions: (1) the model with the highest effect size and (2) whose individual terms are all statistically significant (Supplementary Table 2 for the all the analysis of variance (ANOVA) tables). The only bone whose full model (for example, 'shape ~ $\log_{10}$-body mass + x + y + z') failed to meet the three conditions is the carpometacarpus in which maximum length in the z axis is not statistically significant and therefore, the PGLS model including only maximum lengths in the x and the y axes was used to generate the proportion-normalized shape of this element.

Visual inspection of the residual shapes confirmed our expectations for an approach intended to remove proportional variation and preserve local shape variation (Extended Data Fig. 1 and Supplementary Figs. 1–14).

Quantifying disparity

For all the morphological variables (size proportions, form and shape) and partitions (whole skeleton, regions and individual elements), including both empirical data and our simulations below) we quantified morphological disparity as the mean pairwise squared Euclidean distances among species within the subclade. This was chosen as an estimator of the spread of species in multivariate shape space in favour of other measures, because it has been shown to be robust to variation in sampling proportion and sample size by both theoretical and empirical studies, with properties similar to variance-based measures of disparity[37,60,61].

Patterns of morphospace establishment

To evaluate how extant bird lineages established their morphospace through time, we used ancestral character estimation in the R package mvMORPH v.1.1.4 (ref. [62]) for all our morphological variables (size proportions, form and shape) and partitions (whole skeleton, regions and individual elements), based on principal component axes representing >95% of variation. We then calculated morphological disparity for all

edges of our phylogeny that cut through each in a series of census times, spaced by 1 Ma intervals from the origin of the bird crown group to the present. This provided a time series of the estimated disparity of Neornithes from their time of origin to the present, based on those lineages that are extant today. This was compared to the time series resulting from multivariate Brownian motion simulations (100 iterations, due to the extensive computational time required), using parameters estimated from our empirical datasets, also using functions from mvMORPH[62]. When the empirical line falls above the 95% confidence intervals of the simulated values, that suggests that birds established higher disparity at that time than would be expected under a gradual, diffusive model of evolution. Our approach is similar to that of by Cooney et al.[4], who used univariate simulations of individual principal component axes, also representing >95% of variation. For direct comparison to Cooney et al.[4], we also therefore present analyses using univariate simulations of individual axes based on functions in ape v.5.0 (ref. [63]) and phytools v.1.0-1 (ref. [64]), showing that out multivariate approach is more conservative (that is, has wider 95% confidence intervals, and so is more likely to reject a hypothesis of early establishment of morphospace; see Supplementary Fig. 15 for comparison).

Patterns of morphospace partitioning

To evaluate patterns of morphospace partitioning through time, we calculated morphological disparity for all the subclades in our tree for all our morphological variables (size proportions, form and shape) and partitions (whole skeleton, regions and individual elements). This was expressed as a proportion of the mean disparity of the whole clade (Neornithes). We used these subclade disparities to document the relationship between subclade disparities and subclade ages among birds, and within waterbirds, non-passerine landbirds and passerines, using a modified version of the 'disparity through time' approach[11]. We avoid using this terminology ('disparity through time') because these analyses evaluate morphospace partitioning among progressively younger subclades, differing from the quantification of how disparity accumulated through time in our analysis of morphospace establishment, described above. We therefore call these 'subclade disparity–age' plots, or morphospace partitioning analyses. These provide a test of the mode of evolution, testing the proposed importance of early partitioning of morphospace in birds and bird subgroups.

We summarized subclade disparity as the mean value for all subclades whose stem (branch ending directly in the basal node for that clade) existed at that time. Connecting all these time-binned values creates mean empirical disparity-through-time curves from the origin of a clade to the most recent branching event it experienced (Extended Data Fig. 2).

To establish a macroevolutionary baseline to compare empirical values with, we calculated subclade disparity for 200 simulations of morphological evolution under a uniform, single-rate Brownian motion model (constant diffusive trait evolution) for each kind of morphological data (proportions, form, shape and local shape) and each partition (whole skeleton, regions and individual bones). For this, we used the function mvgls to fit a single-rate Brownian motion model to each kind of data/partition to estimate the adequate trait-covariance matrix and the ancestral states at the root of the tree, then, we fed these values to the function mvSIM, which we used to generate 200 iterations of simulated morphological data. These simulated morphological data represent a diffusive, Brownian motion, model of evolution with a single set of parameters shared across all Neornithes ('all Neornithines' simulations) and was used to evaluate relative disparities of each group across all anatomical data types. We also repeated this process for each subclade (waterbirds, non-passerine landbirds and passerines) by fitting only species data belonging to each of the clades. These were used to evaluate the mode of evolution for each group (that is, early partitioning versus diffusive or constrained evolution; below).

We interrogated the mode of morphological evolution by calculating the MDI[11] for each of the four target groups (Neornithes, Aequolithornithes, non-passerine Inopinaves and Passeriformes). MDI is calculated as the area between the empirical subclade disparity-through-time curve and the disparity-through-time curve simulated under Brownian motion. MDI therefore informs about how disparity among subclades compares with disparity of the total group and at which phylogenetic level constraints in morphological evolution might be acting. Negative MDI values indicate that the empirical curve shows lower subclade disparities than the simulated curve and indicates that subclades explore smaller morphospaces than expected given the diversity of the full clade, suggesting early partitioning of trait space or constrained subclade evolution. Positive MDI indicates that subclades explore larger morphospaces than expected given the diversity of the full clade, suggesting little partitioning of avian trait space, and/or that constraints on evolution at the level of all birds are similar to those of its subgroups[11]. Confidence intervals use the 95 and 5% percentiles of simulated disparities in place of the mean. We calculated MDI using the area.between.curves internal function of the dtt function in the R package geiger.

For display, we calculated the relative subclade disparities compared to expectations under a uniform model of all-bird evolution by subtracting $\log_{10}$-transformed empirical group disparities from mean $\log_{10}$-transformed simulated subclade disparity values from our 'all Neornithines' simulations. We plotted these values as colours in the basal nodes for all clades in our phylogeny to visually inspect patterns of disparity across phylogeny for each data/partition (Fig. 2 and Extended Data Figs. 5–10).

We also summarized patterns of pan-skeletal variation (whole skeletal proportions, allometry-free combined form and proportions-normalized combined shapes) other water-linked groups: Gruiformes (cranes, rails and akin), Charadriiformes (shorebirds), Ardea ('core waterbirds'), Anseriformes (ducks and geese); and all the other land-linked groups that are not members of Inopinaves.

Relative disparity

We compared the relative disparity of our target groups (Aequolithornithes, non-passerine Telluraves and Passeriformes) to expectations under a uniform model across all Neornithes, for all anatomical data types. Relative disparity was calculated as the empirical disparity minus the simulated disparity across 200 Brownian motion simulations ('all Neornithines' simulations). These 200 values were summarized by calculating the minimum, maximum, mean and first and third quantiles. Relative disparities provide information about the difference in a group's disparity compared to what would be expected given the duration and topology of phylogenetic history if all birds evolved in the same way. We repeated this for four more groups: Anseriformes, Galliformes, Gruiformes and a diverse assemblage of non-landbird mainly terrestrial clades (Strisores + Otidimorphae).

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Raw landmark coordinates can be accessed as Supplementary Files linked to this article and accessing our project in OSF Public Repository following this link: https://osf.io/wjk3m/. All three-dimensional meshes can be freely downloaded following the links to Morphosource in Supplementary Materials Table 1.

Code availability

Custom code can be accessed in our project in OSF Public Repository following this link: https://osf.io/wjk3m/.

50. Jetz, W., Thomas, G., Joy, J., Hartmann, K. & Mooers, A. The global diversity of birds in space and time. *Nature* **491**, 444–448 (2012).
51. Bjarnason, A. & Benson, R. A 3D geometric morphometric dataset quantifying skeletal variation in birds. *MorphoMuseuM* **7,** e125 (2021).
52. Abourachid, A., Fabre, A. C., Cornette, R. & Höfling, E. Foot shape in arboreal birds: two morphological patterns for the same pincer-like tool. *J. Anatomy* **231**, 1–11 (2017).
53. R Core Team. *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2019); https://www.R-project.org/
54. Adams, D., Collyer, M., Kaliontzopoulou, A. & Baken, E. Geomorph: Software for geometric morphometric analyses. R package version 4.0.4. https://cran.r-project.org/package=geomorph (2022).
55. Collyer, M., Adams, D. & Collyer, M. M. Package 'RRPP'. RRPP: Linear Model Evaluation with Randomized Residuals in a Permutation Procedure, R package version 1.1.2. https://cran.r-project.org/package=RRPP (2021).
56. Harmon, L. et al. Package 'geiger'. R package version 2 (R Foundation for Statistical Computing, 2020).
57. Bookstein, F. L. in *Advances in Morphometrics* (eds Marcus, L. F. et al.) 131–151 (Springer, 1996).
58. Gunz, P. & Mitteroecker, P. Semilandmarks: a method for quantifying curves and surfaces. *Hystrix* **24**, 103–109 (2013).
59. Collyer, M. L., Davis, M. A. & Adams, D. C. Making heads or tails of combined landmark configurations in geometric morphometric data. *Evolution. Biol.* **47**, 193–205 (2020).
60. Ciampaglio, C. N., Kemp, M. & McShea, D. W. Detecting changes in morphospace occupation patterns in the fossil record: characterization and analysis of measures of disparity. *Paleobiology* **27**, 695–715 (2001).
61. Wills, M. A., Briggs, D. E. & Fortey, R. A. Disparity as an evolutionary index: a comparison of Cambrian and Recent arthropods. *Paleobiology* **20**, 93–130 (1994).
62. Clavel, J., Escarguel, G. & Merceron, G. mvMORPH: an R package for fitting multivariate evolutionary models to morphometric data. *Meth. Ecol. Evol.* **6**, 1311–1319 (2015).
63. Paradis, E. & Schliep, K. ape 5.0: an environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics* **35**, 526–528 (2019).
64. Revell, L. J. phytools: An R package for phylogenetic comparative biology (and other things). *Methods Ecol. Evol.* **3**, 217–223 (2011).

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Author contributions G.N. and R.B.J.B. conceptualized the study and designed the analytical approach. G.N., A.B., E.G. and R.B.J.B. collected and curated the data. G.N. and R.B.J.B. undertook the formal analyses. G.N. and R.B.J.B. wrote the manuscript. G.N. made the figures. G.N., A.B., E.G. and R.B.J.B. edited the text. R.B.J.B. acquired the funding used for this research.

Competing interests The authors declare no competing interests.

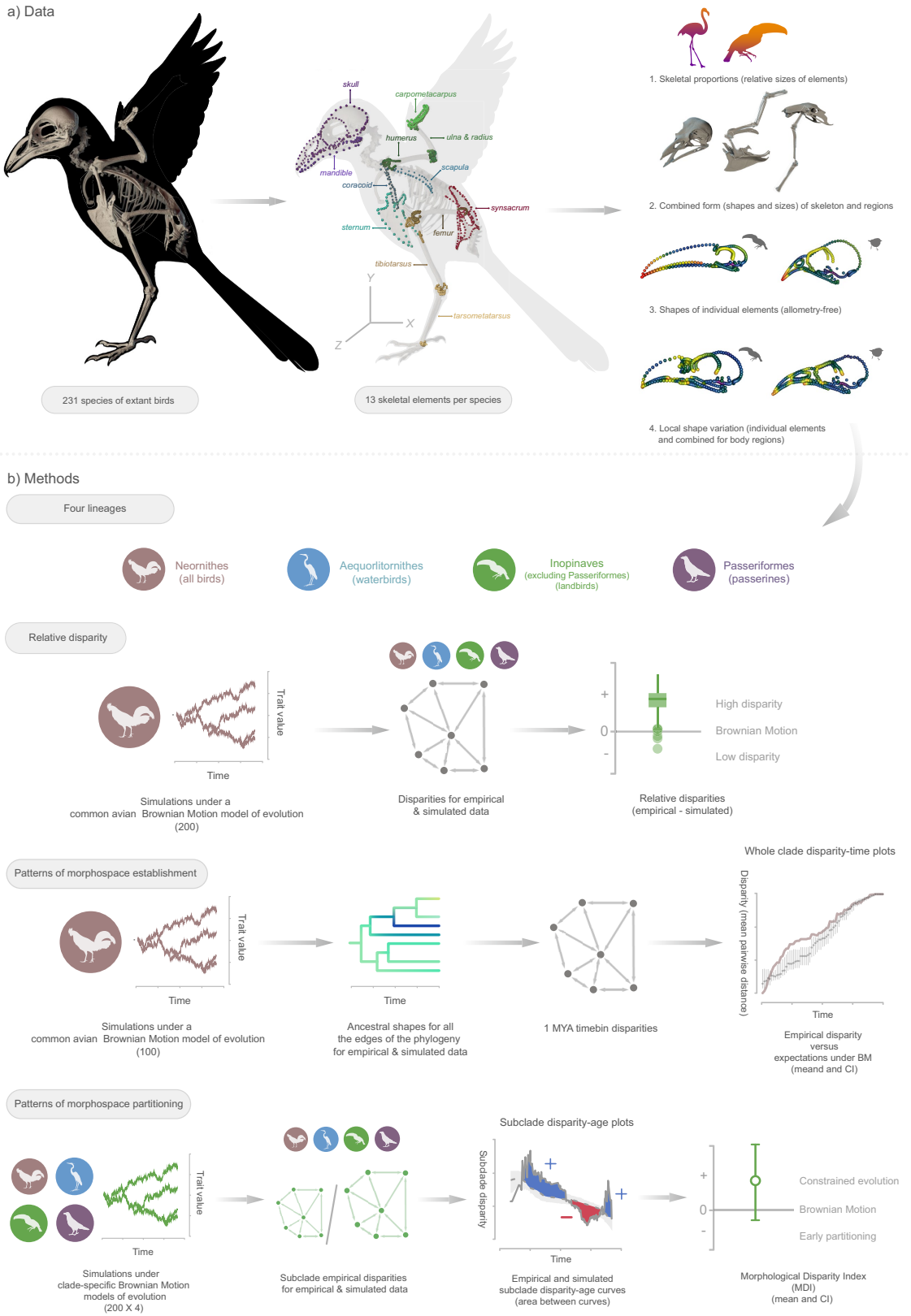
Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-022-05372-y.

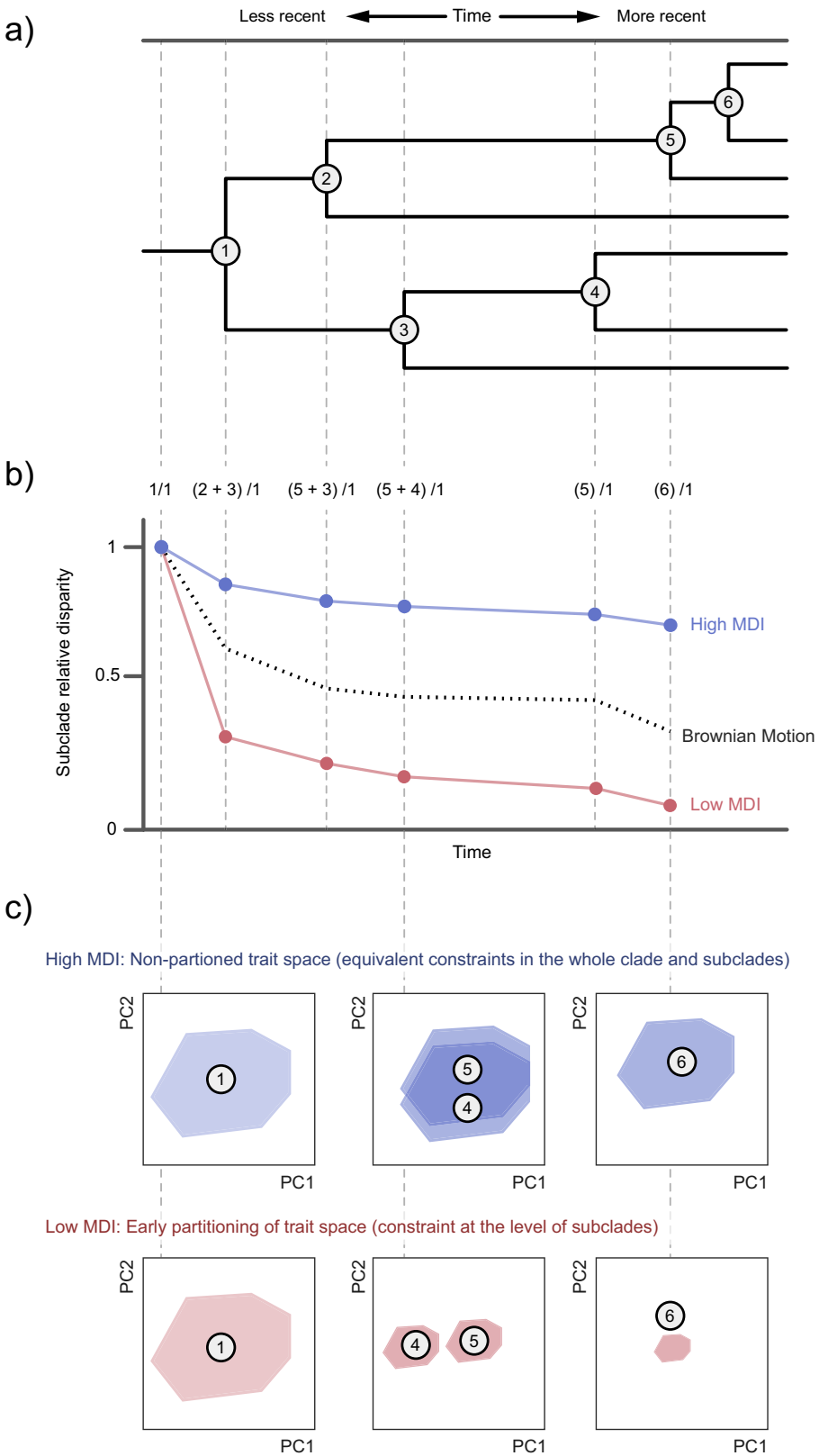
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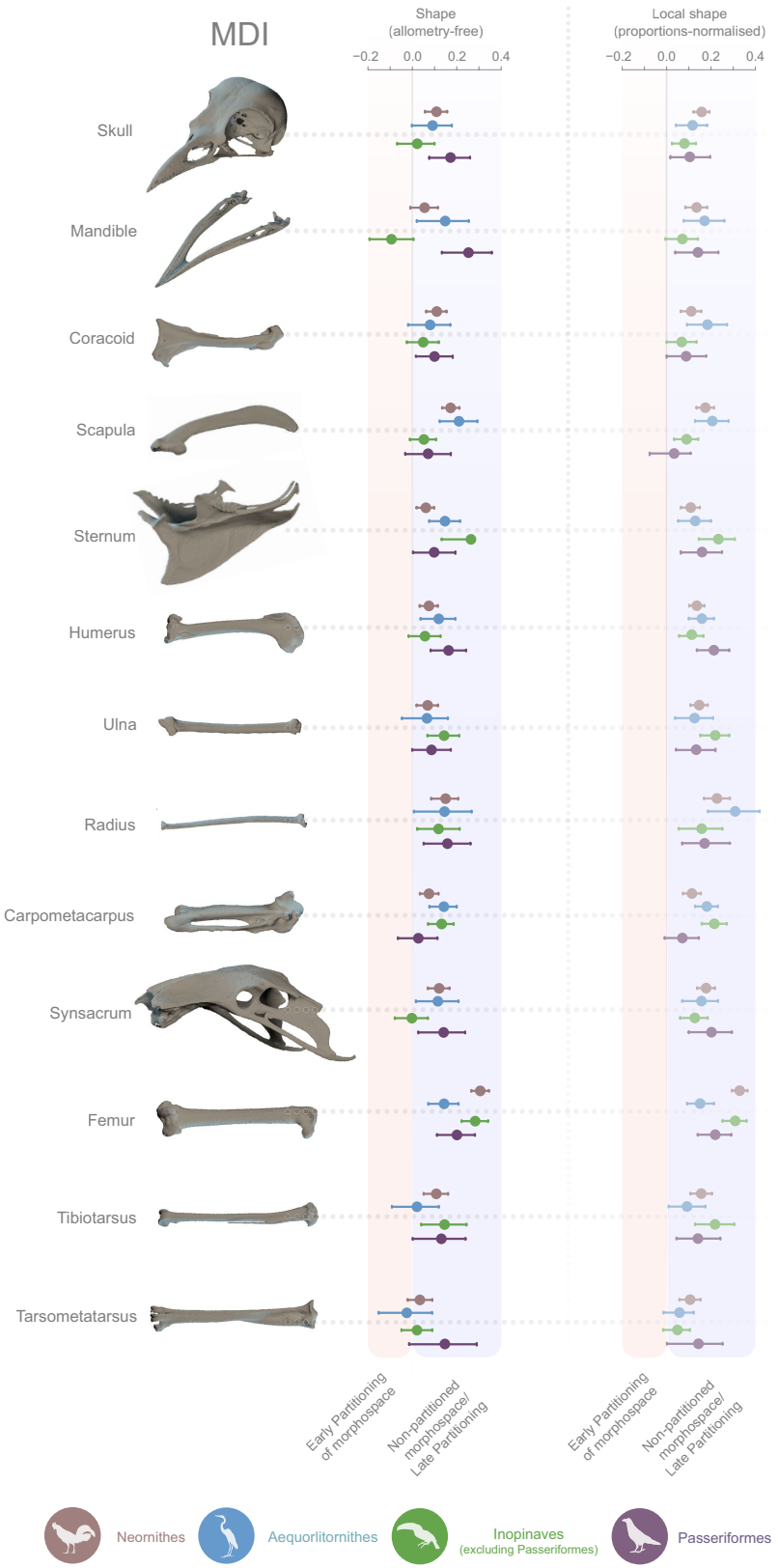
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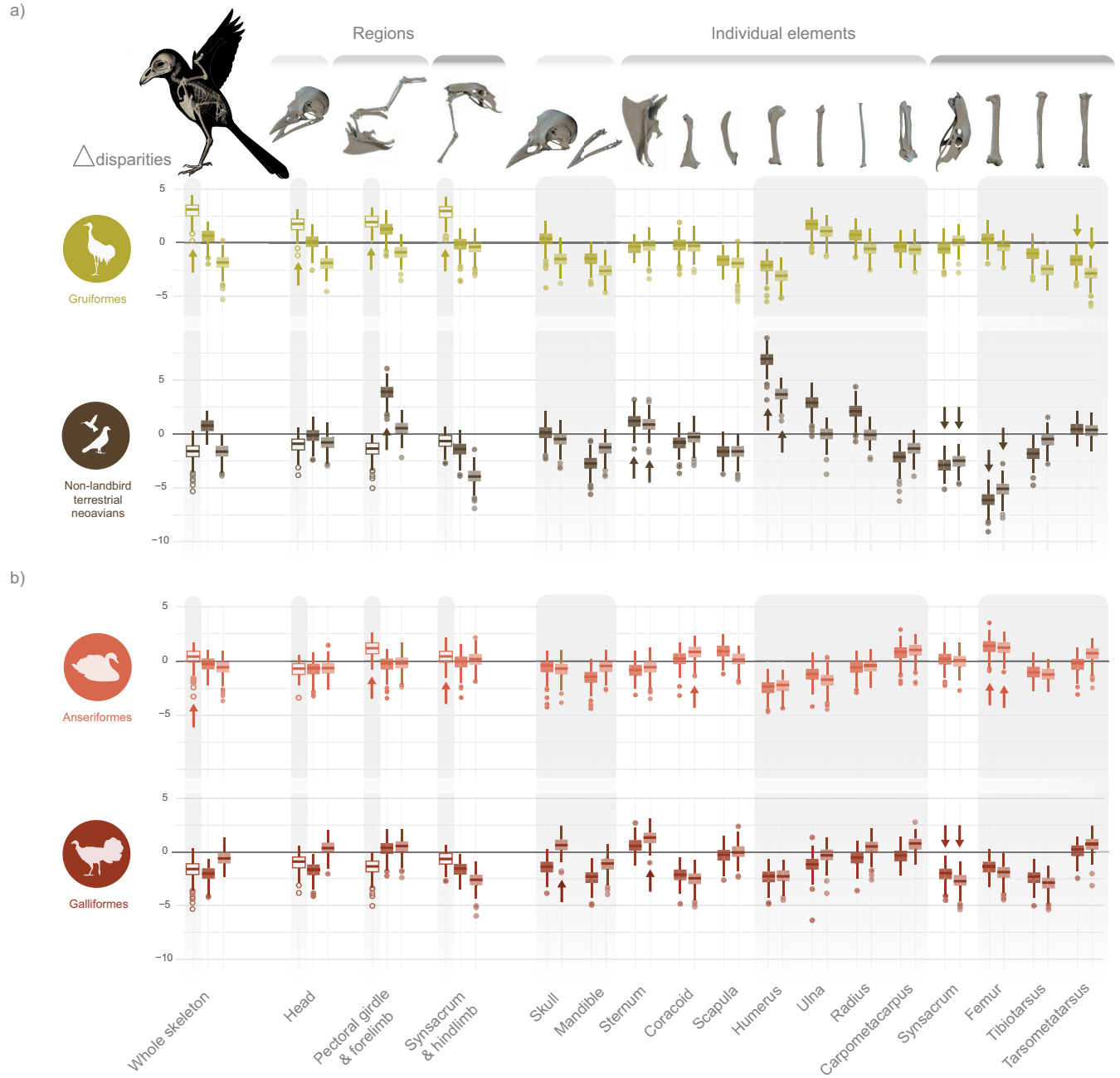
Extended Data Fig. 1 | Infographic displaying the data and the methods' pipeline used in this study. A detailed account of the Material and Methods used in our study can be found in the main text.



Extended Data Fig. 2 | Diagram showing how subclade disparity through time is calculated as test of early partitioning of morphospace, showing idealized contrasting patterns of morphological evolution. a) Example time-calibrated phylogeny and points in time (cladogenesis events) for which b) subclade disparities values are calculated for a Brownian model of diffusive evolution and two alternative modes of morphological evolution. c) Corresponding patterns in morphospace exploration over time for two idealised patterns of morphological evolution: early partitioning of trait space (morphospace) in which morphological evolution is constrained at the level of subclades, and a non-partitioned pattern in which constraints are similar in the total group and within subclades.



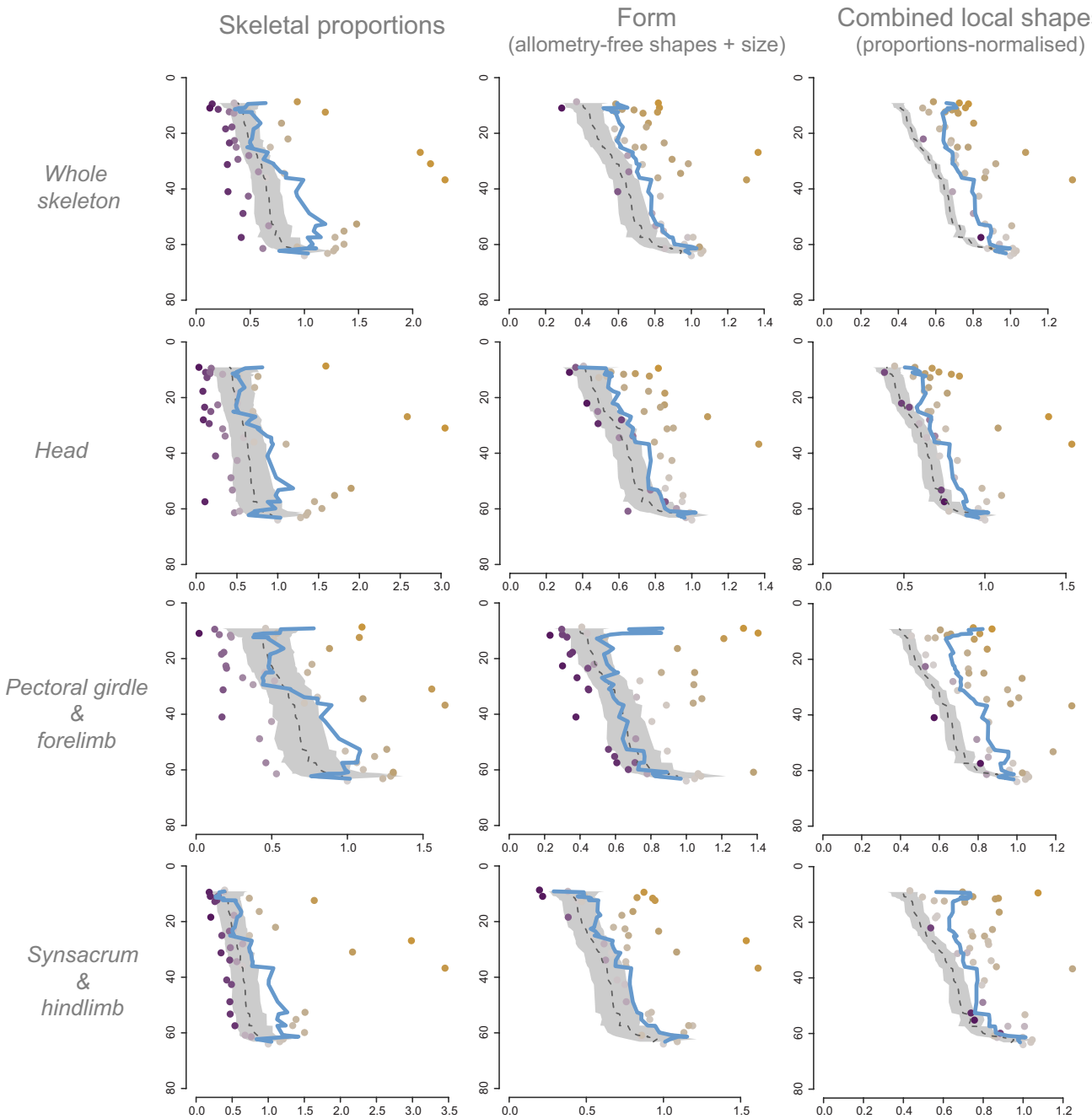
Extended Data Fig. 3 | Tests of early-partitioning of avian allometry-free shape and local shape variation of the individual elements in Aequorlithornithes, Inopinaves and Passeriformes. See Extended Data Fig. 2 for a detailed account of the idealised patterns associated with each mode of evolution of disparity through time.



Extended Data Fig. 4 | Deviations of group-specific disparities from the disparity expected if all birds were evolving under a uniform Brownian motion model of evolution for additional groups of water-based (Gruiformes, Anseriformes) and land-based (non-landbird terrestrial Neoaves, Galliformes) birds. Boxplots summarize the distribution of delta disparities, calculated as empirical disparities minus disparities for the 200 simulated values for each of the three target clades. All partitions and data types are displayed, namely, the whole skeleton, three main skeletal regions and individual bones and the four different aspects of morphological variation in the three target lineages of birds. Arrows highlight the lineage/partition/data type which shows particularly high (up pointing arrows) or low (down pointing arrows) values. Values are normalised by interquartile range.

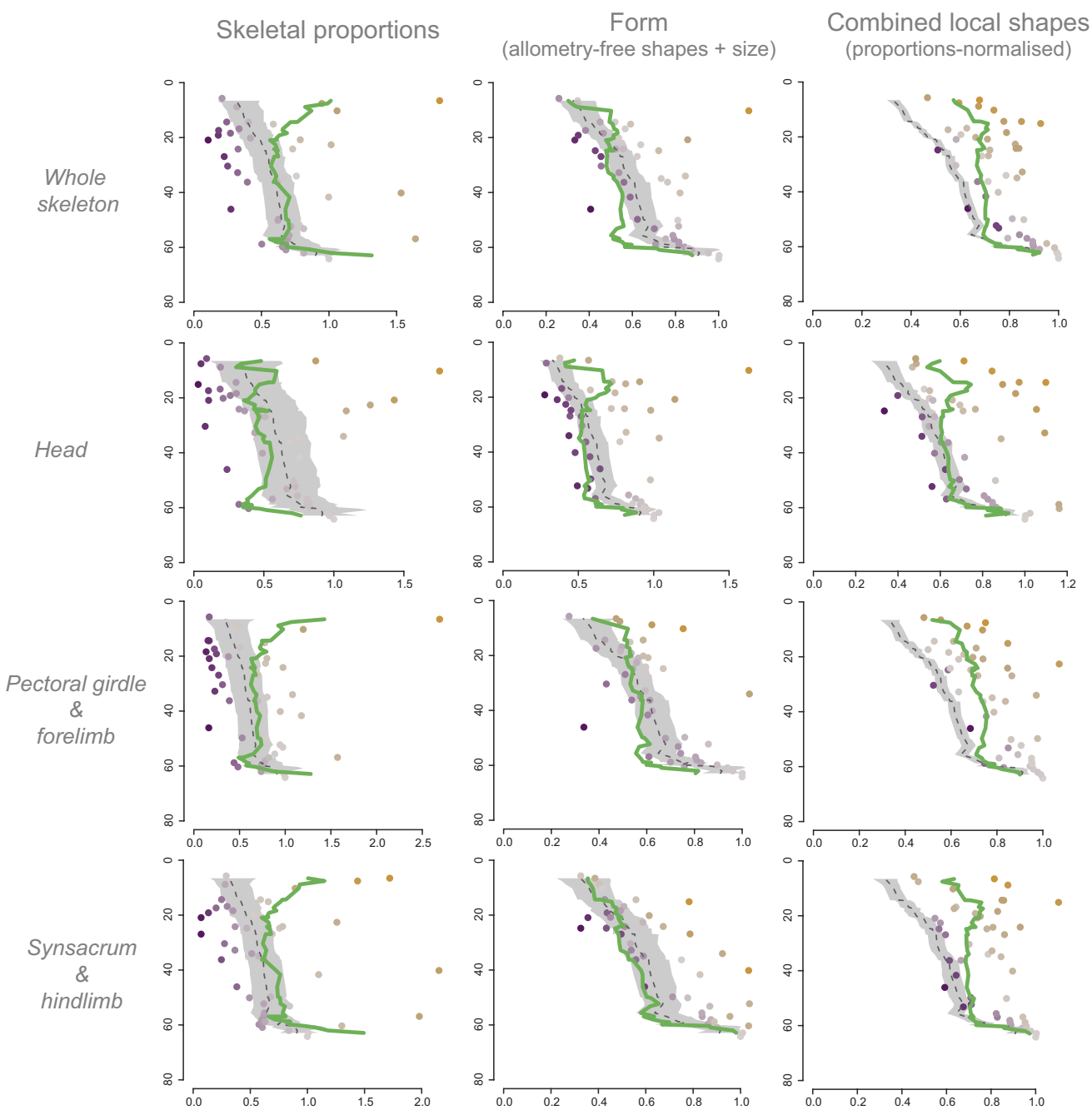


Aequolitornithes (Regions)



Extended Data Fig. 5 | Subclade disparity–age plots for the whole skeleton and body regions for Aequorlitornithes (waterbirds). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.

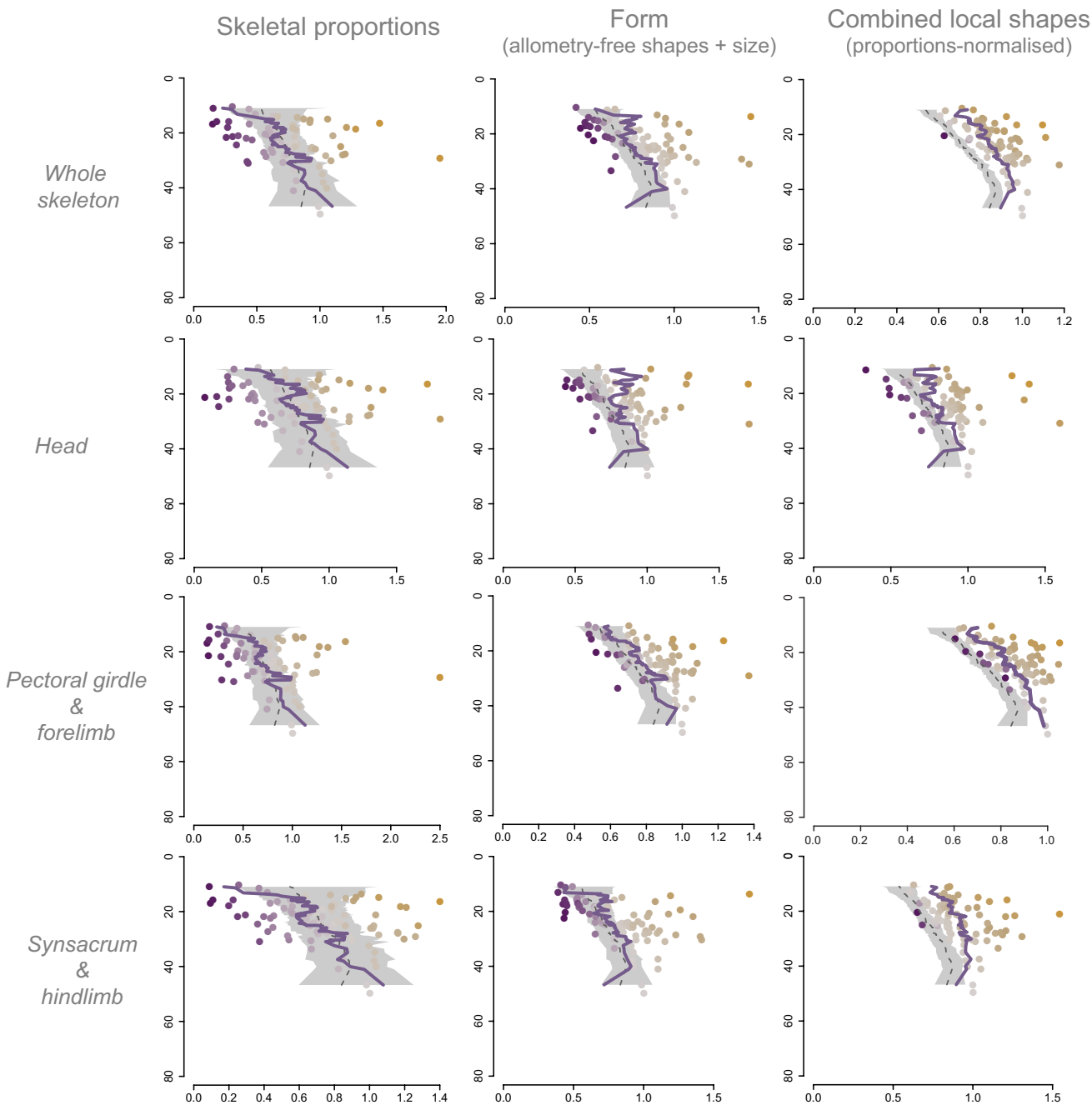




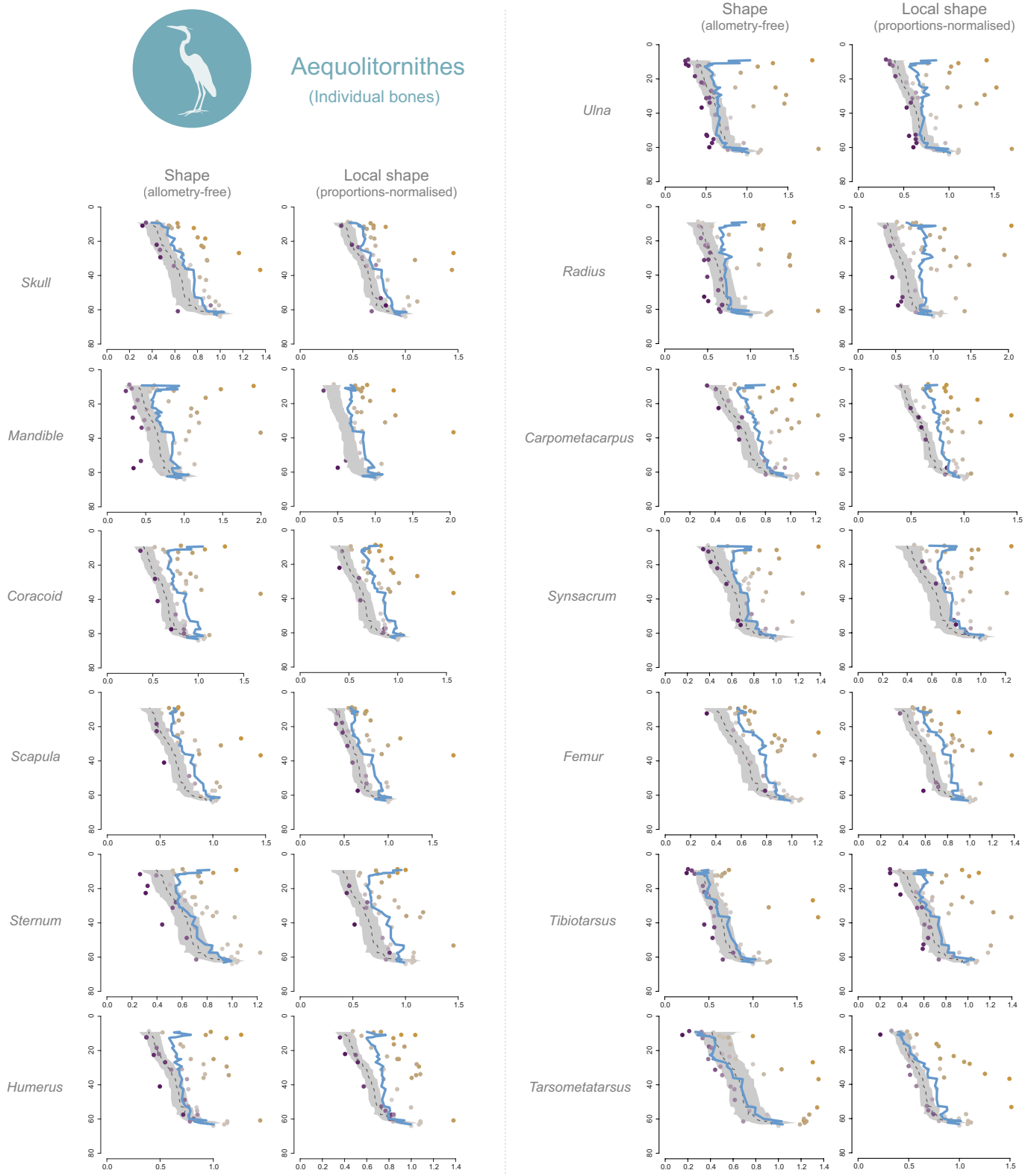
Extended Data Fig. 6 | Subclade disparity–age plots for the whole skeleton and body regions for Inopinaves (landbirds). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.



Passeriformes (Regions)



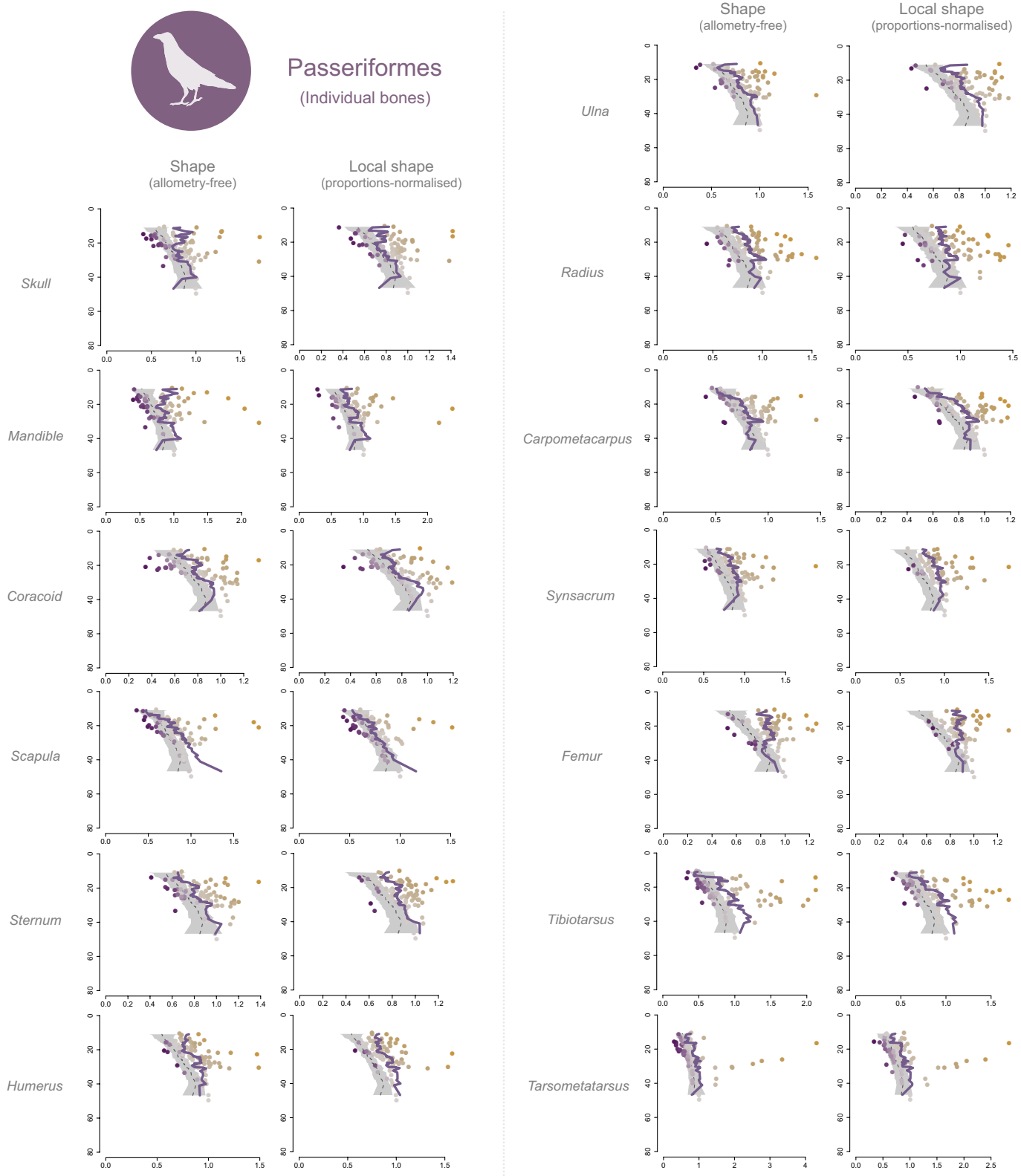
Extended Data Fig. 7 | Subclade disparity–age plots for the whole skeleton and body regions for Passeriformes (passerines). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.



Extended Data Fig. 8 | Subclade disparity–age plots for the individual elements for Aequorlitornithes (waterbirds). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.



Extended Data Fig. 9 | Subclade disparity–age plots for the individual elements for Inopinaves (landbirds). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.



Extended Data Fig. 10 | Subclade disparity–age plots for the individual elements for Passeriformes (passerines). X axes represent subclade disparities while y axes represent time in millions of years from the present time. Solid coloured lines represent mean empirical disparities through time, dashed lines represent mean BM-simulated disparities through time, shaded polygons display the space between 95% percentile and 5% percentiles of disparities through time for BM-simulated data. Colour code of individual subclade disparities (dots) as in Fig. 2.

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Our dataset comprises 3D morphometric data for 13 skeletal units in 228 species, providing a phylogenetically- and ecologically broad sample of extant avian diversity. Specimens were accessioned from several museum collections across the globe (Table S1, specimens used and collection numbers). Digital skeletal models were segmented from high-resolution micro-CT (computed-tomography) scans of complete skeletons in Avizo 9.3. We placed landmarks and curve semilandmarks on these models (total of 207,245 individually placed coordinates), yielding a dense, quantitative representation of each element's geometry and thereby providing a broad representation skeletal form (shape and size) compared to previous studies.
Data analysis	We used a combination of several recent approaches aimed to study different characteristics of the modes of disparity through time, adapted to multivariate data. Our analyses were conducted in R version 4.0.2 using functions primarily from the packages ape version 5.5 , geomorph version 4.0.1, RRPP version 0.6.2, and geiger version 5.3, as well as custom code (Files S3 and S4).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data used can be freely accessed in our OSF Repository (link: https://osf.io/wjk3m/), Morphosource (links in Table S1) and Online Materials in Nature portal. This includes raw landmark data, custom code and original three-dimensional digital meshes for the skeletal elements studied.

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data where this information has been collected, and consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	We studied the patterns of accumulation of skeletal disparity of the living radiation of birds since their origin in the Late Cretaceous to the present time. Critically, we evaluate some predictions of macroevolutionary theory and avian palaeontology concerning the patterns of morphological evolution in the wake of the K-Pg. For this we assembled one of the largest datasets ever assembled for birds and undertook the most comprehensive study of the skeletal phenotype in any group of vertebrates using a combination of geometric morphometrics and phylogenetic comparative methods, including methodological contributions of our own.
Research sample	Our dataset comprises 3D morphometric data for 13 skeletal units in 228 species, providing a phylogenetically and ecologically broad sample of extant avian diversity.
Sampling strategy	Our sampling follows that of the two most comprehensive phylogenomic studies which sampled most lineages of living birds (therefore, being an accurate representation of most deep nodes in the avian tree).
Data collection	Digital skeletal models were segmented from high-resolution micro-CT (computed-tomography) scans of complete skeletons in Avizo 9.3. Thereafter, We placed landmarks and curve semilandmarks on these models, yielding a dense, quantitative representation of each element's geometry. We used this raw information to generate several datasets. We analysed data representing structure at four hierarchical levels of organisation: 1) Overall skeletal form, as the combined relative sizes and allometry-corrected shapes (='forms') of all skeletal elements, or subsets representing the head (skull, mandible), pectoral limb (scapula, coracoid, sternum, humerus, ulna, radius, carpometacarpus), and pelvic limb (synsacrum, femur, tibia, tarsometatarsus) ; 2) Overall skeletal proportions

	represented by a matrix containing the relative sizes of individual elements across the skeleton and for subsets (head, pectoral limb, pelvic limb); 3) The allometry-corrected shapes of individual skeletal elements; and 4) Local shape variation of elements, both individually and combined together per region.
Timing and spatial scale	Not applicable.
Data exclusions	Not applicable.
Reproducibility	All the raw data and code are fully available with the manuscript to allow full reproducibility of our results.
Randomization	Multivariate simulations are used in all the analyses to ensure comparisons against a meaningful macroevolutionary baseline.
Blinding	Not applicable

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging